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A

Power – 12V, solar and USB inputs, 18650 charger, rails

File: power.kicad_sch

ESP32, programming, buttons, RTC, LED strip, buzzer

File: mcu.kicad_sch

RS-485 and uplink

File: comms.kicad_sch

B

AgriNode GW-LAN (LoRa to Ethernet gateway)

One board per hub type: HUB-W (Wi-Fi), HUB-L (LoRa), HUB-C (4G LTE module board), GW-LAN (LoRa to Ethernet)

Core: ESP32-WROOM-32E, 12 V / solar / USB-B inputs, 18650 + TP4056 + power path, MIC29302 3.3 V, DS3231M RTC

USB programming (CH340C, auto-reset) + PROG header, RESET and BOOT/PAIR, WS2813 LED-strip header, buzzer

Optional RS-485 (Modbus) – leave unpopulated if not needed

Source of truth: design/hubs.py – regenerate with build.py <variant>

C

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E

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F

Generated by tools/kigen from the Python design source – edit the source, not this file

AgriNode

Sheet: /

File: agrinode_gw_lan.kicad_sch

Title: AgriNode GW-LAN (LoRa to Ethernet gateway) – Overview

Size: A3

Date: 2026-09-27

Rev: A

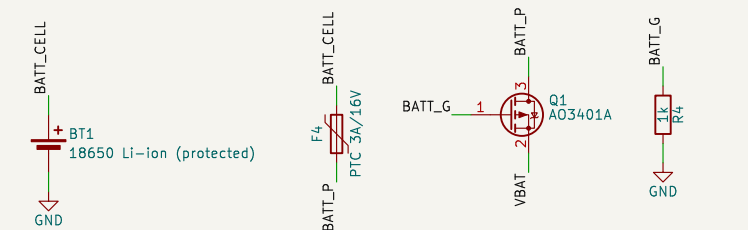
KiCad E.D.A.

kicad-cli 7.0.11+dfsg-1build4

Id: 1/4

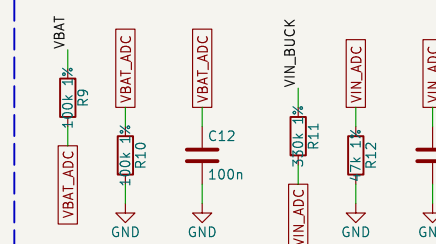
each input fused and reverse-polarity protected; any combination may be connected

18650 cell: holder, fuse, reverse-insertion protection

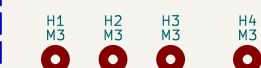


Supply monitoring (ADC1)

VBAT: 100k/100k VIN: 330k/47k (24 V $\rightarrow$ 3.0 V)



Mounting holes (M3)



AgriNode

Title: AgriNode GW-LAN (LoRa to Ethernet gateway) – Power – 12V, solar a

SIZE: AS	DATE: 2025-09-27	ROW: A
KiCad E.D.A.	kiCad-cli 7.0.11+dfsg-1build4	Id: 2/4

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